

Overview of Emmes Timelines for Study Close-Out and Database Lock

The following are typical timelines for study close-out activities leading up to database lock. These are estimates and may vary based on the number of clinical sites and subjects, complexity of the data, available resources, and responsiveness of collaborators. These timelines are intended to facilitate study close-out planning but should be reviewed and agreed upon with the Emmes Clinical Study Manager for each protocol as the study nears last subject last visit.

Finalize Clinical Database

<u>Milestone</u>	<u>Target Time to Completion</u>	<u>Notes</u>
1. Clinical data entry completed	4 days after last subject, last visit	Dependent on site compliance with data entry timelines
2. Data cleaning and querying*	2 weeks after data entry completion; must be complete before final monitoring visit	Dependent on volume of queries and responsiveness to queries; Resolution may require multiple rounds
3. Serious Adverse Event reconciliation	3 weeks after data entry completion; must be complete before final monitoring visit	Assumes one round of review and reconciliation
4. Final monitoring visit(s)	5 weeks after last subject, last visit	Dependent on data cleaning and querying progress, SAE reconciliation, and availability of site staff and monitor
5. Resolution of all data management and monitoring queries	1 day after final monitoring visit	Assumes no new data management queries generated as a result of monitoring activities
6. Clinical database lock	1 day after resolution of all data management and monitoring queries	Dependent on completion of Items 1-5

*This step has several components; see page 2.

Finalize Core Laboratory/Immunology Data**

<u>Milestone</u>	<u>Target Time to Completion</u>	<u>Notes</u>
1. Assay completion and transfer of laboratory data to Emmes	12 weeks after last specimen collection time point	Protocol-specific , dependent on assay; Assumes results format has been previously communicated to Emmes.
2. QC of laboratory data and distribution of queries to the laboratory	4 days after receipt of laboratory data	Dependent on data being delivered in agreed upon format
3. Resolution of laboratory queries	1 week after distribution of queries	Dependent on volume of queries and responsiveness to queries; Resolution may require multiple rounds
4. Laboratory database lock	1 day after resolution of queries	Dependent on completion of Items 1-3

**Bioinformatics and some exploratory immunology data may not follow this timeline; see page 3.

Data Cleaning and Querying Activities:

Data cleaning and querying encompasses multiple components that are related to one another – as one activity is completed, it may trigger or impact another component. These are described below but there may be additional steps depending on the specific protocol.

- **At a single site:**

- *Completion of data entry*
 - All electronic Case Report Forms (eCRFs) are entered into Advantage eClinical or the Emmes Data Manager (DM) has granted requests for an exception to submitting the eCRF.
 - All required questions within an eCRF are completed or the Emmes DM has granted requests for an exception to submitting the question.
 - All research specimens are entered into GlobalTrace and shipped to the final destination (DMID CMS, other repository or testing laboratory).
 - All specimens have been entered into GlobalTrace and discrepancies (e.g., missing specimens, unexpected specimens) are resolved.
 - All Adverse Events are resolved per the protocol.
 - All concomitant medications have an end date entered per the protocol.
 - All abnormal laboratory values have the relationship to treatment ascribed.
 - All pregnancies are completed (the database may be partially locked while an ongoing pregnancy is followed).
- *Resolution of data queries*
 - All automated data queries (e.g., data anomalies on the study website or automated data queries in Advantage eClinical) are resolved.
 - All manual queries in Advantage eClinical are resolved.
 - All GlobalTrace discrepancies (e.g., collection date discrepancies, and future use discrepancies) are resolved.
- *Review of completed data*
 - The Emmes Clinical Safety and Pharmacovigilance Subject Matter Experts (SMEs) have reviewed all adverse events and medical history for coding according to the Medical Dictionary for Regulatory Activities.
 - The Emmes Clinical Safety and Pharmacovigilance SMEs have reviewed all concomitant medications for coding according to the WHODrug Dictionary.
 - The Emmes Clinical Data Interchange Standards Consortium (CDISC) SME has reviewed the data to ensure it conforms to the standards for mapping to Study Data Tabulation Model (SDTM) format.
 - All data submitted to Advantage eClinical have been evaluated by the site monitor or CRA according to the Clinical Monitoring Plan and there are no outstanding discrepancies.
 - The site PI has completed attestation in Advantage eClinical that the data entered into the eCRFs are complete and accurate.
 - The Emmes Biostatistician has reviewed the data to ensure it is sufficient to fulfill the requirements for primary protocol endpoints, safety assessments, and any other key end points.

- **Across all sites:**

- The Statistical Analysis Plan (SAP) is final and approved.
- The analysis populations for the analyses described in the SAP are final and approved.
- CDISC SDTM mapping is complete and has been independently reviewed for quality control.

Post-Database Lock Reporting Activities:

In addition to the Interim Clinical Study Report, there are several other reporting activities that occur toward study end. These reporting activities may follow independent timelines or may be dependent on finalization of the Interim CSR.

- **Clinical Study Report (Interim)**

- Once the clinical and non-clinical (laboratory) databases are locked, Emmes generates the tables, figures, and listings (TFLs) for the CSR.
- Emmes provides TFLs and supporting materials to DMID-CROMS approximately 8 weeks after database lock.
- The CSR compilation processes and timelines are detailed within the DMID CSR Compilation Guidelines document.

- **Addendum to the Clinical Study Report**

- Bioinformatics analyses and some exploratory endpoints may take significantly longer to generate the data than the conduct of the trial itself as well as the primary/secondary assays. In this situation, DMID may approve the completion of a primary CSR to include the safety and primary/secondary laboratory data and allow for the bioinformatics and/or exploratory assay data to be submitted as an addendum to the CSR once the data are available.

- **Final Safety Monitoring Committee/Data and Safety Monitoring Board (SMC/DSMB) Report**

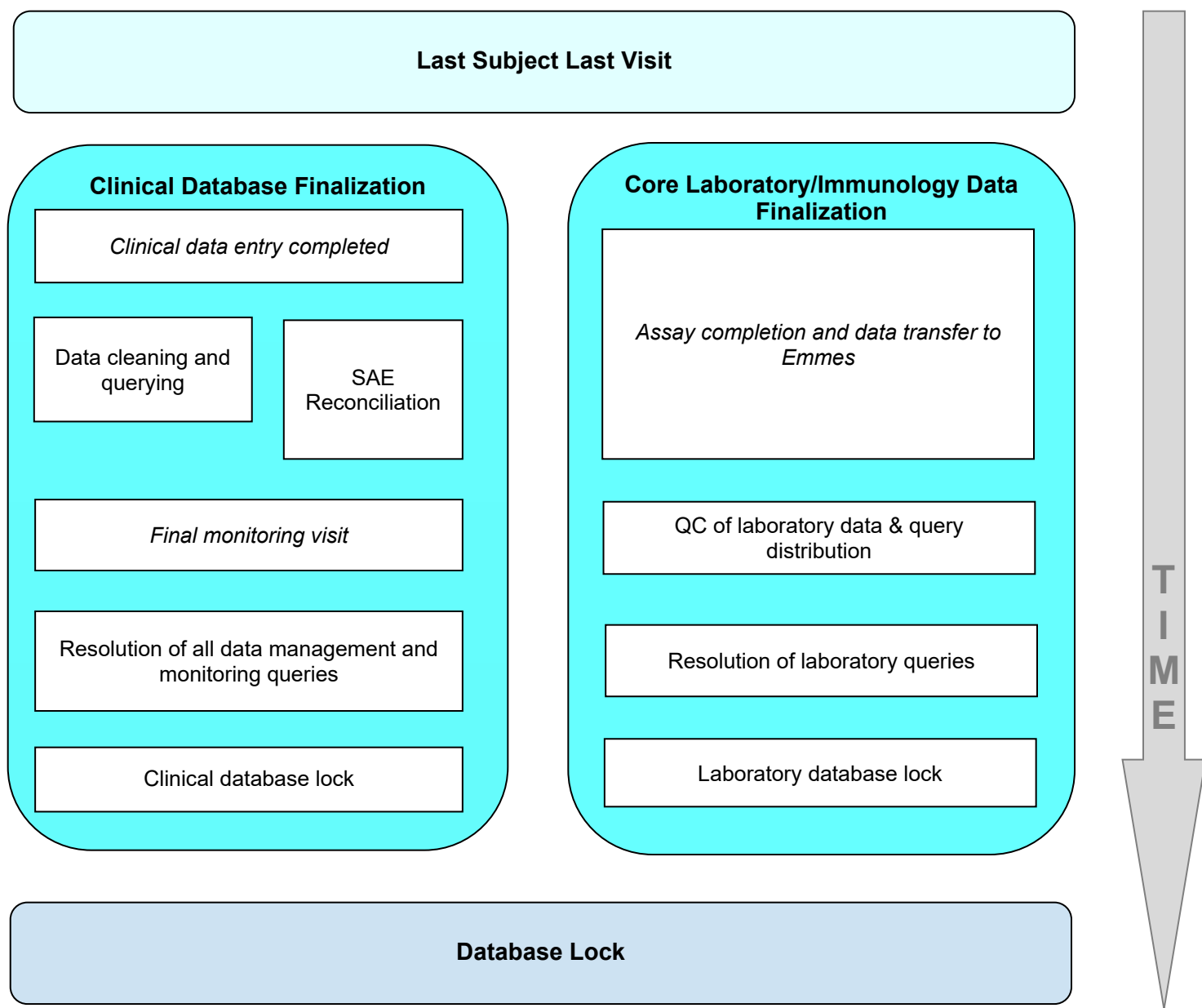
- A final report is provided to the SMC/DSMB after the CSR is submitted to the FDA.
- As directed by DMID, the full CSR or a subset of the TFLs from the CSR may be provided to the SMC/DSMB to meet the final reporting needs.

- **National Library of Medicine (NLM) Results Reporting**

- The timeline for NLM Results Reporting is based on the completion of the **primary endpoint(s)**. Results for **at least one** primary endpoint must be uploaded to NLM within one year of the date primary endpoint data are **collected from the last subject**.
 - If reportable endpoints are based on a laboratory test, the deadline is one year after the last specimen collection for that test.
 - If reportable endpoints are based on a safety parameter (e.g., SAEs), the deadline is one year after that last safety assessment is conducted.
- Note that the time point for the reportable endpoints may be prior to last subject, last visit and is not dependent on database lock. Therefore, it is possible that NLM reporting may occur prior to finalization of the CSR.
- Emmes will distribute tables to the PI for review approximately 7 days prior to the due date for submission of the results record to DMART, which is due 2 weeks prior to the 1-year deadline. The review is primarily as a courtesy, as minimal text is allowed in the record, and interpretative text is strictly forbidden by NLM.
- Currently, DMID determines if a protocol meets the reporting requirements to NLM.

- **Manuscript Writing**

- Manuscript writing may begin once the data are locked (or even before, with the results section pending database lock and analysis); however, the manuscript may not be submitted for publication until after the CSR has been submitted to the FDA.
- Manuscripts may only be submitted prior to submission of the CSR to the FDA if it is specified in the protocol.



Italics are indicative of non-Emmes responsibilities